

Parth Doshi

(510) 641-7070 | pmdoshi@ucsc.edu | linkedin.com/in/parthmdoshi | github.com/PartyD1

EDUCATION

University of California, Santa Cruz

Bachelor of Science in Computer Science – GPA: 4.00

Santa Cruz, CA

Expected: December 2028

- **Relevant Coursework:** OOP, Systems, Data Structures, ML Basics, Linear Algebra, Assembly, C, UNIX/LINUX

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, HTML, CSS, Bash, Tailwind, Elixir

Frameworks/Libraries: FastAPI, Flask, React, Next.js, PyTorch, TensorFlow, NumPy, Pandas, Pydantic, LangChain

Developer Tools: Node.js, PostgreSQL, Supabase, OpenAI API, NemoClaw, Vercel, Apify, Git, Jupyter

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern

InstaService

June 2026 - August 2026

San Jose, CA

- Built a **multi-modal agentic framework** that helped minimize operations staff workflow time **by up to 70%**.
- Implemented continuous 0–100 risk scoring across six weighted signals, surfacing silently-failing bookings and converting the process into a ranked queue, resulting in faster identification of risky bookings by approximately **25%**.
- Orchestrated three specialized agents: risk detection, provider tracking, and chat monitoring, evaluating every booking in parallel, enabling coordinated responses that accelerated handling of risky bookings by up to **40%**.
- Enabled reliable autonomous operation, sending SMS, calls, & emails, alerting operations staff of emergency situations, and automatically escalating bookings, mitigating impediments by up to **90%** fully autonomously.

Undergraduate Research Assistant – Augmented Design Lab

University of California, Santa Cruz (Advisor: Prof. Jim Whitehead)

April 2026 – Present

Santa Cruz, CA

- Engineered a domain-specific LLM agent with LangChain (Node.js) using structured API tool-calling, achieving **99.2% accuracy** on live NBA event queries by grounding responses in real-time data rather than model memory.
- Implemented a tile-based procedural generator in TypeScript using the Wave Function Collapse algorithm and enforced adjacency constraints, consistently producing valid map layouts and eliminated adjacency violations.
- Reverse engineered **3+** open-source WFC implementations and research papers to reconstruct constraint propagation logic, reducing invalid tile configurations and improving generation reliability across edge-case inputs.
- Modeled map generation using **SAT/SMT** representations, eliminating edge-case failures in generation pipelines.
- Synthesizing **20+** research papers across PCG, artificial intelligence, and agent-assisted content generation systems.

PROJECTS

Santa Claws | Python, Next.js, NemoClaw, OpenClaw, Nemotron, Supabase

May 2026

- Deployed autonomous agent orchestration in **24 hours** by implementing NemoClaw and **OpenClaw**, NVIDIA's alpha-stage sandboxed runtime, with heartbeat execution on Nemotron 3 Nano Omni 30B via NVIDIA Brev GPUs.
- Architected a Supabase/PostgreSQL-driven orchestration layer as the queue, shared memory, and audit log with durable row-claiming and 30–1200s heartbeat cycles, reducing complexity without sacrificing fault tolerance.
- Accelerated a full sales pipeline from lead discovery to cold emailing by engineering **4 specialized agents** with **20 Python tools** spanning scraping, website scoring, mockup generation, email critique loops, and cold-emailing.
- Scaled outreach to **500 leads in ~8 hours** by integrating **4+ external APIs** into an autonomous pipeline.

ScorelyAI – AI Report Grading Platform | FastAPI, PostgreSQL, Next.js, OpenAI API

February 2026

- Enabled **30+ active users in the first month** and partnered with a high school to provide **150+ students** with access, achieving structured deployment for 2027, demonstrating adoption at scale in an educational setting.
- Automated grading of **100+ student reports** in under **20 seconds per report**, processing **20,000 tokens per report**, reducing human grading time by over **400%** while maintaining **2.8%-point average deviation**.
- Implemented a robust async FastAPI & PostgreSQL pipeline for PDF ingestion, schema-validated GPT-4o prompt responses via Pydantic, and event-specific grading logic, achieving section-level feedback for **16+ event types**.
- Employed computer vision for compliance checks, creating an identification accuracy of **99.4%** for visual errors.